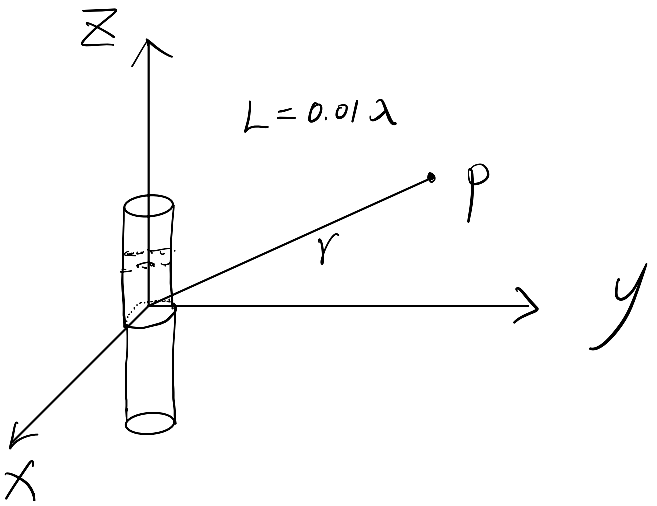
Q1 [redacted]

The dipole is as follows:



a) at point P, magnetic vector potential $\vec{A} = \vec{a}_z L I \frac{e^{-jkr}}{4\pi r}$

$$\vec{H} = \frac{1}{\mu}\nabla \times \vec{A} \approx \vec{a}_\phi jk \frac{LI}{4\pi r} e^{-jkr} \sin\theta \quad A/m$$

$$\vec{E} = \nabla \times \frac{\vec{H}}{j\omega\varepsilon} \approx \vec{a}_\theta j\omega\mu \frac{LI}{4\pi r} e^{-jkr} \sin\theta \quad V/m$$

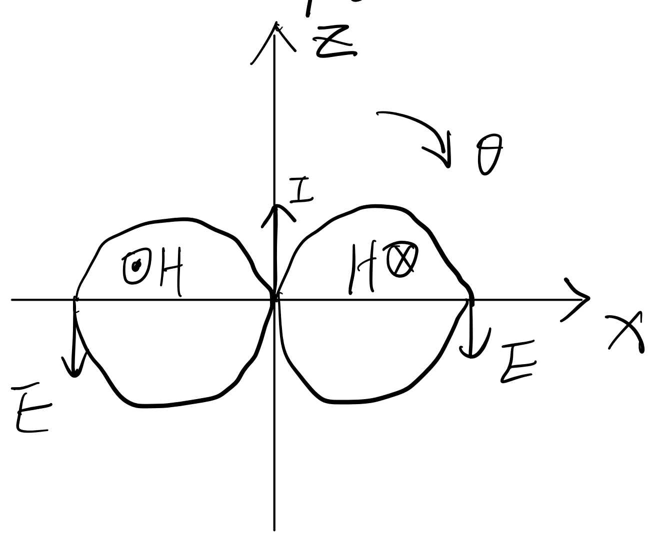
$\because L = \frac{\lambda}{100}$, $k = \frac{2\pi}{\lambda} = \omega\sqrt{\mu\varepsilon}$, $\therefore \vec{H} = \vec{a}_\phi j \frac{I}{200r} e^{-j\frac{2\pi r}{\lambda}} \sin\theta$

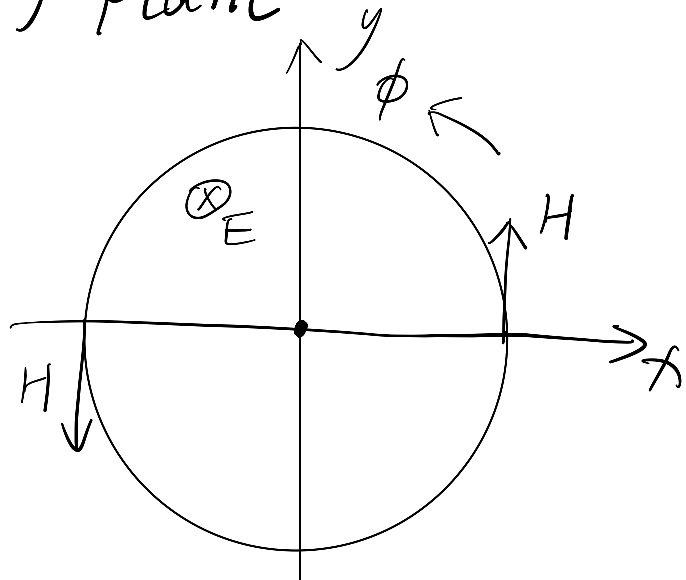
$\because \sqrt{\frac{\mu}{\varepsilon}} = \eta$ $\therefore$ $\vec{E} = \vec{a}_\theta j\sqrt{\frac{\mu}{\varepsilon}} \cdot \frac{I}{200r} e^{-j\frac{2\pi r}{\lambda}} = \vec{a}_\theta j \frac{\eta I}{200r} e^{-j\frac{2\pi r}{\lambda}}$

$E_\phi = E_r = H_\theta = H_r = 0$

b) when current I along $\vec{z}$

in x-z plane   in x-y plane





c)

complex power density =

$$\vec{S} = \frac{1}{2}\vec{E}\times\vec{H}^* = \frac{1}{8}\eta\left(\frac{IL}{\lambda}\right)^2 \frac{\sin^2\theta}{r^2}\,\vec{a}_r \quad W/m^2$$

$$= \frac{\eta I^2}{80000}\cdot\frac{\sin^2\theta}{r^2}\,\vec{a}_r$$

rediated power =

$$P_{rad} = \frac{\pi}{3}\eta\left|I\frac{L}{\lambda}\right|^2 = \frac{\pi\eta}{30000}I^2 \quad W$$

in free space, $\eta = 120\pi = 377\,\Omega$,

$$\vec{S} = 4.71\times10^{-3}\left[\frac{I\sin\theta}{r}\right]^2\vec{a}_r,\quad P_{rad} = 0.0395\,I^2$$

d)

$$R_{rad} = \frac{2P_{rad}}{|I|^2} = \frac{\pi\eta}{15000} = 0.079\,\Omega$$

e) Radiation Intensity =

$$U(\theta,\phi) = S\cdot r^2 = \frac{1}{8}\eta\left(\frac{IL}{\lambda}\right)^2\sin^2\theta$$

$$V_{ave} = \frac{P_{rad}}{4\pi}$$

Directivity =

$$D = \frac{V(\theta,\phi)}{V_{ave}} = \frac{\frac{\eta I^2L^2\sin^2\theta}{8\lambda^2}\times4\pi}{\frac{\pi\eta I^2L^2}{3\lambda^2}} = \frac{3}{2}\sin^2\theta$$

$$D_{max} = \frac{U_{max}}{V_{ave}} = \frac{3}{2}$$

$\because R_{loss} = 1\,\Omega \quad \therefore e_{rad} = \frac{R_{rad}}{R_{loss}+R_{rad}} = \frac{0.079}{1+0.079}$

$= 0.073$

gain =

$G = e_{rad} D = 0.11 \sin^2\theta$

$G_{max} = e_{rad} D_{max} = 0.11 = -9.59\ dB$

f) radiation efficiency $= e_{rad} = 0.073$

$|S_{11}|_{dB} = 20\log|\Gamma| = -10, \quad |\Gamma| = 0.316$

matching efficiency =

$e_m = 1 - |\Gamma|^2 = 0.9$

antenna efficiency =

$e_{ant} = e_m e_{rad} = 0.0657$

realized gain =

$G = e_{ant} D_{max} = 0.0657 \times \frac{3}{2} = 0.0986$

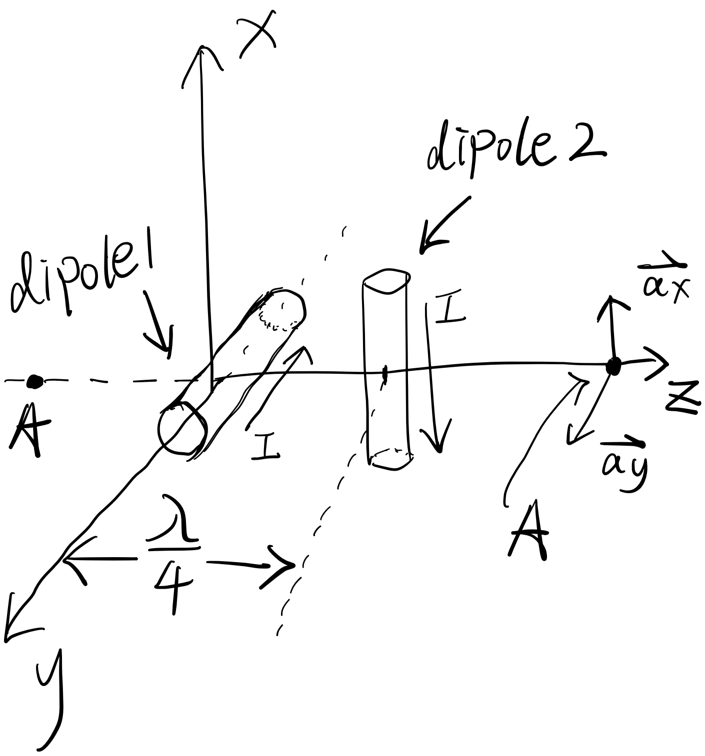
g)

$L = 0.01\lambda < 1.25\lambda,$

$R_{ff} = 3\lambda$

Q2

a) Here is the antenna system.



I set point A on both two side of z-axis, the distance between A and origin point is r.

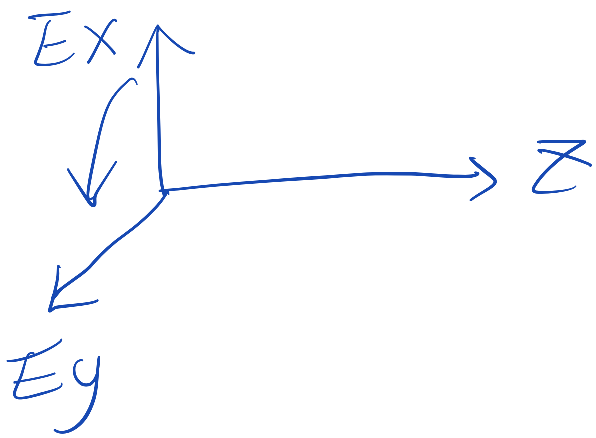
The direction/arrow of I is the initial current direction. (t=0)

I put the center of dipole 2 on the z-axis along the x-axis. Two dipoles are vertical

Dipole 1 generates E_y and Dipole 2 generates E_x.

The max magnitude of E_x and E_y are same, meet the condition of circular polarization.

Both dipoles generate E at the same time, I put dipole 2 on $z=\frac{\lambda}{4}$, so dipole 2 will be ahead of dipole 1. Because A will receive E_x from dipole 2 firstly, then receive E_y. The phase gap is $\frac{\pi}{2}$, which meet the condition of circular polarization.



b) when $Z_A = r$, $\because r \gg \frac{\lambda}{4}$ $\therefore$ the max magnitude of E_x and E_y are same equal to E_0. I set Dipole1 generates $\vec{E_y} = \vec{a_y} E_0 \cos(wt - kz)$

Then $\vec{E_x}$ generated by dipole 2 should be

$$\vec{E_x} = \vec{a_x} E_0 \cos(wt - k(z - \frac{\lambda}{4}))$$

$$= \vec{a_x} E_0 \cos(wt - kz + \frac{\pi}{2})$$

Total $\vec{E}$ at $Z_A = r$:

$$\vec{E} = \vec{a_x} E_0 \cos(wt - kr + \frac{\pi}{2}) + \vec{a_y} E_0 \cos(wt - kr)$$

c) when $Z_A = -r$, Because the E_y generated by dipole1 on the positive and negative Z-axis is symmetrical.

$\vec{E_y}$ should be $\vec{E_y} = \vec{a_y} E_0 \cos(wt - k|z|)$

$$= \vec{a_y} E_0 \cos(wt + kz) \quad (z < 0)$$

when $Z_A = -r$,

$$\vec{E_y} = \vec{a_y} E_0 \cos(wt - kr)$$

$\vec{E_x}$ will change to

$$\vec{E_x} = \vec{a_x} E_0 \cos\left(\omega t - k\overbrace{\left(|z| + \frac{\lambda}{4}\right)}^{\text{distance between dipole 2 and A}}\right)$$

$$= \vec{a_x} E_0 \cos\left(\omega t - k|z| - \frac{\pi}{2}\right)$$

$$= \vec{a_x} E_0 \cos\left(\omega t + kz - \frac{\pi}{2}\right) \quad (z<0)$$

Because the value after k is the distance between dipole and point A.

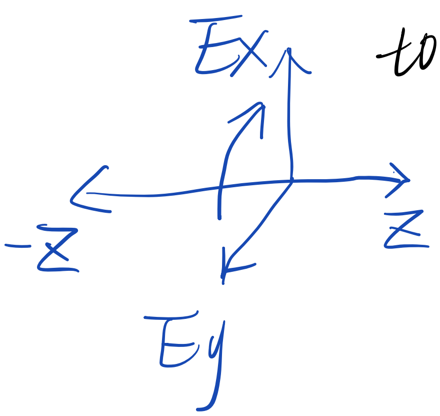
when $z_A = -r$,

$$\vec{E_x} = \vec{a_x} E_0 \cos\left(\omega t - kr - \frac{\pi}{2}\right)$$

Total E:

$$\vec{E} = \vec{a_x} E_0 \cos\left(\omega t - kr - \frac{\pi}{2}\right) + \vec{a_y} E_0 \cos(\omega t - kr)$$

d) on the negative z-axis, the direction of radiation is "-z". The phase of E_y generated by dipole 1 will be ahead of E_x, because physically dipole 1 is close to negative A than dipole 2, and from c), we can calculate phase $E_y - E_x = \frac{\pi}{2}$, so it's still RHCP



Q3

Using two orthogonal antennas as the transmitter, two independent channels can increase the capacity of transmission. Besides, when this system is used as receiver, the interference between two paths can be reduced, and the receiving efficiency can be improved

In mobile communication base-station, when we use horizontal - vertical antenna to polarize EM waves, the horizontal component of EM wave will be reduced due to the loss of the ground/earth induction when the EM wave propagate close to the ground. In that case, the vertical and horizontal EM components will increase imbalance, and ±45° polarization can reduce this case

In addition, in practice especially in the city, when the orthogonal components of EM wave meet obstacles, they will reflect and refract, V-H polarization waves have higher correlation between two components than ±45°, so ±45° polarization can avoid coupling and confusion better, and it can improve the channel independence and have better transmission capacity.